

Decision Authorization for Trustworthy AI: Semantic Institutional Admissibility in a Neurosymbolic Control Plane

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Abstract

Language models and tool-using agents increasingly produce outputs that can influence institutional decisions, invoke external tools, and trigger downstream workflows. Existing work has advanced retrieval, agent capability, safety evaluation, access control, and system-level control, yet a distinct deployment question remains: *when is this exact AI-mediated movement institutionally admissible to become a protected consequence under the conditions that govern it now?* We study that question as *decision authorization*. The central distinction is between capability, structural permission, and semantic institutional admissibility. A system may be able to act, and a route may be technically permitted, while the proposed movement is still unsupported, unauthorized, contextually invalid, or too consequential for autonomous release.

This paper presents a public-safe systems architecture for that boundary. In the present implementation, the outward decision contract is bounded to ALLOW, ESCALATE, or BLOCK. Materially incomplete or authority-insufficient authorization inputs fail closed to ESCALATE, withholding autonomous release and routing unresolved judgment to review while preserving honest missing-evidence and missing-authority posture. A separate pre-transition research direction studies whether a candidate has become sufficiently formed to enter authorization at all; that non-formation concept is not presented here as a fourth public outcome or as current live Decision API behavior. The conceptual architecture distinguishes semantic decision responsibility from deployment enforcement responsibility; it does not claim that the current private implementation realizes those responsibilities as physically separate runtime components. It further requires observable evidence about identity, authority, current governing context, review posture, release status, consequence/no-bind state, and artifact integrity.

A private implementation, OntoGuard, provides one bounded *controlled engineering snapshot* showing that the outward contract is executable. In the archived 5 September 2026 snapshot, a five-case batch produced one ALLOW, three ESCALATE, and one BLOCK outcome, while the unchanged engineering conformance harness for that exact artifact set recorded 2,501 passing assertions with no warnings or hard failures. The assertion count is a snapshot-specific historical observation, not a standing score, product rating, cumulative metric, or public benchmark sample count; later harness counts are not implied by this paper. These results are engineering evidence, not public benchmark superiority or production non-bypassability claims. The paper contributes the problem definition, conceptual architecture, observable invariants, comparison with adjacent control layers, testable hypotheses, falsifiers, and an implementation-independent evaluation protocol. Implementation algorithms, formulas, thresholds, rule logic, private schemas, authority-resolution mechanics, lifecycle ordering, longitudinal methods, feedback activation mechanisms, and deployment-specific enforcement topology remain outside the public disclosure boundary.

Reproducibility and Disclosure Scope

The central claims in this paper are intentionally framed so that they can be evaluated from observable inputs, outputs, evidence claims, and controlled consequences without access to the private implementation. A public evaluator should be able to test whether the present system preserves candidate identity, emits a single bounded decision, fails closed rather than autonomously allowing materially incomplete or authority-insufficient cases, represents missing evidence and authority honestly, refuses stale or mismatched authorization, and avoids overstating consequence or enforcement claims. Pre-transition non-formation is treated separately as a research direction rather than as a current fourth public decision outcome.

The paper therefore specifies *what a decision-authorization system must make testable*, not the private mechanism by which OntoGuard computes those results. Detailed disclosure boundaries are summarized in Appendix A. This structure is intended to preserve scientific falsifiability while avoiding an implementation recipe.

1 Introduction

The most important question for a capable AI system is not always whether it can produce a plausible answer or invoke an available tool. In institutional settings, a stronger question often comes next: *is this exact proposed movement admissible to become an organizational consequence under the conditions that govern it now?*

Retrieval-augmented generation (RAG) improves access to external information [1–3], while graph-oriented approaches add structured context and multi-hop evidence relationships [4]. Agent research extends language models from text generation toward planning and tool-mediated action [5, 6]. ToolEmu, AgentDojo, InjecAgent, and tau-bench demonstrate operational failure modes in tool-capable agents, including unsafe actions, prompt-injection vulnerabilities, policy failures, and inconsistent user-facing outcomes [7–10]. Classical access-control research formalizes structural authorization over actors, roles, permissions, operations, and objects [23]. AI-control research asks how system-level mechanisms can constrain unsafe behavior even when model intent is untrusted [11].

These lines of work are complementary, but they do not collapse into one problem. We call the remaining gap the *institutional admissibility gap*. A tool can be structurally available while a particular use of it is not institutionally authorized. A response can be factually plausible while its proposed consequence exceeds delegated authority. A policy citation can be relevant but insufficient for the specific movement. An agent can follow a technically valid route while the route is semantically wrong for the governed intent, current evidence, consequence class, or institutional conditions.

The key mismatch is therefore between *what the system can do*, *what the infrastructure permits*, and *what the institution can legitimately stand behind*. This paper treats that third question as a separate systems problem: *decision authorization*. The research object is not merely an answer, retrieval result, or API call. It is a candidate movement toward a protected consequence. Decision authorization asks whether the movement is sufficiently formed and current to be evaluated, and if so, whether that exact movement is admissible to proceed.

1.1 Research Contributions

This paper makes seven public-safe contributions:

1. It defines **semantic institutional admissibility** as a control problem distinct from capability, retrieval quality, generic tool permissioning, and post-hoc monitoring.

2. It distinguishes the **current fail-closed authorization behavior from a separate pre-transition formation question**: live materially incomplete or authority-insufficient authorization inputs are withheld through ESCALATE, while non-formation remains a distinct research direction rather than a current fourth public outcome.
3. It defines a **bounded decision-authorization contract** and a set of observable invariants that can be tested independently of the private algorithm.
4. It distinguishes, at the **conceptual responsibility level**, semantic decision authority from deterministic enforcement, so future or external deployment controls can consume the bounded semantic result without implying that the current private implementation is physically split into separate decision and enforcement services.
5. It defines **consequence honesty**: a decision is not itself proof that a protected effect did or did not occur, and production non-bypassability requires deployment-specific evidence.
6. It extends the research program from individual events to **observable behavioral governance over time** while requiring longitudinal signals and future-governance feedback to remain non-decisioning unless separately authorized.
7. It provides **testable hypotheses, falsifiers, a claim-to-test matrix, and bounded implementation evidence** without presenting private engineering conformance as benchmark superiority.

1.2 A Running Intuition: Same Capability, Different Admissibility

The institutional-admissibility problem becomes clearest when the technical capability stays constant while the governing context changes. Table 1 gives illustrative pairs. **These examples are not published decision rules.** They are conceptual matched-context examples and do not disclose live corridor prompts, private trigger logic, thresholds, or OntoGuard decision rules.

Technical capability	Context A	Context B	Why the governance question changes
Send a message	Internal drafting support	Customer-facing recommendation with material consequences	Release surface, delegated authority, and consequence differ.
Update a record	Add a reviewer note	Commit an adverse institutional determination	The second movement can create a protected effect.
Clinical assistance	Summarize information for a licensed reviewer	Deliver patient-facing treatment instruction autonomously	Reviewer role and consequence authority differ.
Hiring workflow	Summarize candidate materials	Produce an autonomous ranking that drives a protected employment decision	The downstream effect and required institutional authority differ.

Table 1: Illustrative matched-capability pairs. The technical mechanism can be similar while institutional admissibility differs. These examples define evaluation structure, not private decision logic.

2 Related Work and the Missing Boundary

2.1 Retrieval and Structured Evidence

RAG combines parametric generation with retrieved external information [1]. Subsequent work has explored retrieval critique, graph-based context, and broader RAG design choices [2–4]. These methods improve the information available to a model, but evidence availability is not itself authorization. A source can be relevant yet non-authoritative for the proposed action, or authoritative in general but inapplicable to the current release surface or consequence.

2.2 Agents, Tools, and Operational Risk

ReAct and AgentBench demonstrate the capabilities and evaluation challenges of interactive language-model agents [5, 6]. ToolEmu, AgentDojo, InjecAgent, and tau-bench expose operational, security, and policy risks when language models use tools or interact with users under domain rules [7–10]. Decision authorization complements this literature by asking whether a particular proposed institutional movement should become a protected consequence even when the technical capability to attempt it exists.

2.3 Structural Authorization and AI Control

Classical role-based access control associates permissions with roles, users, operations, and objects [23]. Modern policy and gateway systems extend structural authorization in many directions, but their central task remains evaluating rules that are already encoded in the authorization policy. Decision authorization addresses a different layer: whether the semantic content and institutional meaning of the particular AI-mediated movement are admissible under current evidence, authority, task, release, and consequence conditions.

AI-control research studies system-level mechanisms for constraining unsafe behavior even under adversarial or untrusted model behavior [11]. That work is highly relevant to the enforcement and control problem. The present paper focuses on a complementary question: what semantic determination should be supplied to an enforcement mechanism before protected commitment?

2.4 Model Evaluation, Truthfulness, Uncertainty, and Assurance

TruthfulQA and HaluEval evaluate truthfulness and hallucination behavior [12, 13], while HELM emphasizes broad multi-metric evaluation [14]. Uncertainty research examines model self-awareness and calibration [15, 16]. Neurosymbolic research combines learned representations with symbolic structures to improve interpretability and controllability [17]. Safety-assurance work such as AMLAS and evaluations-based safety cases emphasizes structured evidence for system-level claims [18, 19]. Decision authorization uses these concerns as evidence inputs or assurance context but addresses the runtime admissibility of a specific proposed movement.

2.5 Governance Frameworks

NIST’s AI Risk Management Framework and Generative AI Profile emphasize governance, measurement, risk identification, and trustworthy deployment [20, 21]. The European Union AI Act establishes legal obligations for specified AI systems within a risk-based regulatory structure [22]. Decision authorization is not a substitute for legal or governance frameworks; it is an engineering abstraction for binding a particular AI-mediated movement to the evidence, authority, release, review, and consequence claims that apply at runtime.

2.6 Positioning Relative to Adjacent Control Layers

Table 2 summarizes the typical emphasis of adjacent system classes. The table is intentionally non-exclusive: real systems can overlap. The point is to isolate the research question that decision authorization makes primary.

System class	Typical primary question	Primary object	Relationship to institutional admissibility
RAG / graph retrieval	What evidence or context should inform generation?	Information retrieval and grounding	Supplies evidence; does not by itself authorize consequence.
Agent runtime	What action or tool can the model plan or invoke?	Capability and orchestration	Makes action possible; admissibility remains a separate question.
Access-control / policy layer	Is this sub-object/action/resource structurally permitted under encoded policy?	Deterministic authorization	Can enforce structural rules; semantic institutional meaning must already be represented in policy or supplied by another layer.
AI control / safety evaluation	How can harmful or untrusted behavior be detected, constrained, or evaluated?	System safety and control	Complementary; may consume or enforce semantic judgments.
Assurance case / governance framework	What evidence supports a broader system or governance claim?	Assurance and organizational governance	Provides higher-level evidence and obligations; not ordinarily a per-movement runtime semantic decision.
Decision authorization	May this exact AI-mediated movement become a protected consequence under current institutional conditions?	Semantic institutional admissibility	Makes that question primary and produces a bounded decision for downstream consumption.

Table 2: Typical emphasis of adjacent controls. The categories can overlap; the contribution is the explicit semantic-admissibility boundary.

3 Problem Definition: Capability Is Not Institutional Admissibility

Consider an AI system that produces a candidate output capable of influencing an external workflow. Three questions must be separated:

1. **Capability:** Can the system produce the output or invoke the technical mechanism?
2. **Structural permission:** Is the tool, route, API, or workflow technically permitted by surrounding infrastructure?
3. **Institutional admissibility:** Under the current evidence, authority, scope, policy, intent, review, and consequence conditions, may this particular movement legitimately form and proceed?

The third question is the subject of this paper. The same action shape may be acceptable in one context and inadmissible in another. The same technically available route may be acceptable

for non-binding internal support but unacceptable for a protected customer-facing, clinical, legal, financial, safety, privacy, security, or employment consequence. A previously acceptable movement may also require fresh authorization after a material change in governing conditions.

3.1 Protected Consequence

A *protected consequence* is a downstream institutional effect for which the organization requires explicit governance before commitment. The category is intentionally open-world. It is not defined by a fixed industry list; the relevant consequence and governance context depend on the current task and attached institutional conditions.

3.2 Current Fail-Closed Incompleteness and the Pre-Transition Research Direction

The present OntoGuard implementation does not expose non-formation as a fourth public Decision API outcome. When an authorization request reaches the current decision boundary with materially incomplete decision input or insufficient required authority, the system fails closed to ESCALATE: autonomous release is withheld, unresolved judgment is routed to review, and missing evidence or authority remains explicit rather than being fabricated as sufficient.

A separate pre-transition research direction asks an earlier question: whether a candidate has become sufficiently formed to enter authorization at all. In that research abstraction, non-formation is an eligibility concept rather than ALLOW, ESCALATE, or BLOCK. The paper does not claim that this pre-transition non-formation semantics is current live Decision API behavior; it should be treated as a future or separately validated capability until implemented and independently evidenced.

3.3 Testable Research Hypotheses

The public scientific thesis can be evaluated through the following hypotheses:

1. **H1—Context sensitivity.** For matched technical actions, materially different evidence, authority, release surface, or consequence conditions should produce different authorization outcomes when expert labels differ.
2. **H2—Boundary value.** Adding semantic decision authorization to an otherwise identical agent/control stack should reduce unsafe autonomous release relative to structural permission alone, without degenerating into universal blocking or universal escalation.
3. **H3—Identity integrity.** Authorization for one candidate, route, subject, or governing context should not be silently reusable for a materially different one.
4. **H4—Fail-closed insufficiency.** In the present implementation, materially incomplete or authority-insufficient authorization inputs should not yield autonomous ALLOW. They should fail closed to ESCALATE, with release withheld and missing evidence or authority represented honestly. A pre-transition non-formation state may be evaluated separately in future work, but it is not assumed to be a current fourth public outcome.
5. **H5—Decision non-interference.** Reporting, explanation, projection, and descriptive longitudinal analytics should not alter the canonical current-event decision.
6. **H6—Consequence calibration.** Claims that a protected effect did not form should be made only when consequence evidence supports that statement; a semantic BLOCK alone is insufficient.

7. **H7—Prospective learning.** Later outcomes may inform future governance recommendations without rewriting historical decisions or silently modifying current authority.

4 Public-Safe Architecture of a Semantic Decision Control Plane

The public architecture is best described by responsibilities rather than implementation sequence. Figure 1 is therefore conceptual: it shows the control responsibilities that must compose, not the internal architecture of OntoGuard.

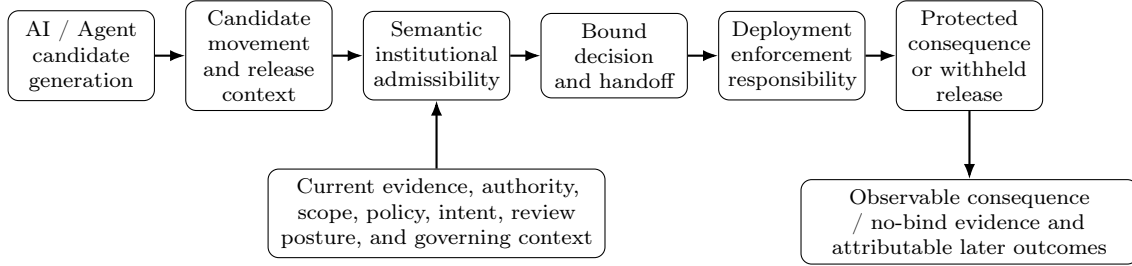


Figure 1: Conceptual responsibility boundary. The figure is not a diagram of the current private repository or a claim that decisioning, packaging, handoff, and enforcement are deployed as separate runtime components. It deliberately does not disclose OntoGuard implementation sequence, data structures, scoring, or internal control logic.

4.1 Candidate and Intent Binding

The system must identify the AI-mediated candidate under review and bind the authorization record to the intended task, subject, release context, and movement being evaluated. This prevents a decision about one output, action, route, or context from being silently reused for another. For agentic systems, the governed object can include an observable proposed action rather than only generated text.

4.2 Neurosymbolic Evidence Mediation and Open-World Scope

The research abstraction is neurosymbolic: learned semantic representations can be combined with symbolic or ontology-mediated structures so that evidence, concepts, scope, and institutional context become traceable governance inputs. This layer can improve the linkage between a candidate movement and the evidence or concepts used to evaluate it, but it is not itself authorization.

The framework is also open-world and provider-neutral. It should not require a fixed list of industries, model vendors, agent frameworks, tool protocols, or enforcement products. New domains and workflows should be representable through current context and evidence rather than hard-coded scenario templates.

4.3 Current Incompleteness Handling and Future Formation Eligibility

In the present live OntoGuard contract, candidates that reach the Decision API receive one of the three public outcomes: ALLOW, ESCALATE, or BLOCK. Materially incomplete or authority-insufficient inputs do not become a separate public eligibility state; they fail closed to ESCALATE, autonomous release is withheld, and missing evidence or authority remains explicit. The paper therefore does not claim that current live OntoGuard first emits a separate formation or eligibility disposition before producing its three-way decision.

A distinct pre-transition research direction may later evaluate whether a candidate is sufficiently formed to enter authorization at all. In that research abstraction, non-formation is an internal eligibility concept rather than a fourth public decision outcome. Until that capability is separately implemented and independently validated, it is treated here as future research rather than present product behavior.

4.4 Semantic Decision Authority

For an authorization-eligible movement, the semantic decision layer emits exactly one bounded outcome:

- **ALLOW**—the scoped movement is authorized for the specified release surface under the current governing conditions;
- **ESCALATE**—autonomous release is withheld and additional qualified judgment or evidence is required; or
- **BLOCK**—the formed movement is not admissible for the scoped release surface in its current form.

The public contract requires a single semantic decision authority. Explanatory analytics, reports, dashboards, human-readable projections, and longitudinal observations must not become parallel decision systems.

4.5 Evidence, Authority, and Governing Context

An authorization record should expose whether the decision is supported by current evidence and whether relevant authority is attached, partial, unavailable, or explicitly not asserted. It should identify the scope and release context for which the decision is valid. The system must not fabricate legal authority, reviewer authority, policy authority, evidence, citation precision, or outcome closure to make the packet appear complete.

4.6 Conceptual Semantic Handoff and Enforcement Responsibility

The paper distinguishes two responsibilities conceptually: determining whether a movement is institutionally admissible, and enforcing or withholding a protected effect in a deployment. The current private implementation should not be inferred from this distinction to consist of physically separate runtime services; decisioning and packet production may be co-located within one implementation pipeline.

In deployment architectures that separate these responsibilities, a downstream control—native to the deployment or supplied by an external policy/workflow system—can consume the bounded semantic result before protected commitment. A public-safe handoff should therefore be sufficiently bound to the movement and current authorization context that such a consumer can reject stale, mismatched, or invalid authorization. The exact handoff structure and validation mechanism are outside the public disclosure boundary. The scientific distinction is functional: structural permission and semantic institutional admissibility answer different questions even when one implementation co-locates multiple responsibilities.

4.7 Consequence and No-Bind Evidence

When autonomous release is withheld, the system should record what can be truthfully established about downstream consequence. In a controlled seam this can include evidence that a protected effect did not form. In a shadow or post-hoc setting, the system must not imply that it prevented an event unless enforcement evidence exists.

A BLOCK result is therefore not equivalent to proof of non-bypassability. The former is a semantic decision; the latter is a deployment claim requiring evidence about actual enforcement behavior.

4.8 Human Review and Outcome Closure

ESCALATE is a governed outcome, not an error state. A useful system should preserve the unresolved judgment and enough supporting evidence for a qualified reviewer to act. Reviewer qualification, authority, outcome, and closure must remain unasserted until attributable evidence exists. Later outcomes can inform future governance, but they should not rewrite the historical decision that was actually made.

4.9 Public Design Invariants

The architecture is defined by seven externally testable invariants:

1. **Exact-movement binding:** the decision is about the movement actually presented for authorization.
2. **Current-context validity:** materially changed governing conditions require fresh evaluation.
3. **Single decision truth:** outward projections cannot manufacture or rewrite the canonical decision.
4. **Non-fabrication:** missing evidence, authority, review, or consequence remains explicitly missing.
5. **Decision/enforcement separation:** semantic authorization can be consumed by, but is not conflated with, deterministic enforcement.
6. **Consequence honesty:** claims about protected effects are no stronger than the available consequence evidence.
7. **Historical immutability:** later observations or future-governance recommendations do not rewrite prior authorization truth.

5 Beyond Single Events: A Broader Semantic Governance Program

Decision authorization is the center of a broader semantic-control research program. Table 3 describes the public role of the principal capability families without exposing their private algorithms or data models.

5.1 Governing-Context Continuity

An authorization should not remain current merely because its historical integrity can still be verified. Material changes in movement identity, evidence, authority, route, policy, task, or other governing conditions can require a fresh authorization. Scientifically, this is a temporal-validity problem: integrity of an old decision is not equivalent to present authority.

5.2 Task-Intent Continuity

For continuing agent or workflow tasks, observable proposed actions can be evaluated against the governed task rather than treated as unrelated events. This is distinct from hidden chain-of-thought monitoring: only bounded observable actions or explicitly supplied task information are in scope.

Capability family	Public research question	Decision-authority boundary
Pre-transition formation research	May a candidate become sufficiently defined to enter authorization at all?	Separate research direction; non-formation is not claimed as a current fourth public Decision API outcome.
Governing-context continuity	Is a previously valid authorization still current after material change?	Can require fresh evaluation; does not create a second decision system.
Task-intent continuity	Do observable actions remain faithful to the governed continuing task?	Supplies decision-relevant or explanatory evidence under bounded use; no hidden reasoning is required.
Observable trace governance	What governance-relevant behavior is supported by attached action, route, tool, review, commit, or outcome traces?	Evidence is bounded by observability; absent traces cannot be inferred.
Longitudinal governance	What descriptive patterns appear across comparable governed episodes?	Descriptive and non-decisioning for the current event.
Governed feedback	What verified outcomes suggest future governance review or change?	Recommendation only until separately authorized prospectively.

Table 3: Public research breadth. The table states scientific roles and authority boundaries, not internal implementation mechanics.

5.3 Observable Trace Governance

A semantic control plane can consume bounded, provenance-bearing observable traces in pre-commit or post-hoc modes. Relevant observations may include action requests, tool use, route selection, retries, review events, commits, non-commit events, and verified outcomes when actually available. The framework must remain explicit about partial observability.

5.4 Bounded Longitudinal Governance

Across sufficiently comparable governed episodes, longitudinal analysis can describe recurring review dependence, evidence deterioration, authority pressure, route pressure, repeated refusal attempts, or changing consequence severity. Such analytics are observational unless stronger causal evidence exists. They must not silently alter the current event’s ALLOW, ESCALATE, or BLOCK result.

5.5 Governed Feedback

Verified reviewer and downstream outcomes can support recommendations for future governance improvement. A recommendation is not current authority and may not autonomously alter model weights, policy, ontology, retrieval behavior, delegated authority, route configuration, thresholds, or historical decisions. Any future governance change must be separately authorized and must apply prospectively to a fresh decision.

6 Observable Decision Authorization Contract

The central scientific artifact is the Decision Authorization Packet. Publicly, the packet should be understood as a contract of observable claims rather than a published schema.

Claim family	Publicly observable requirement
Governed candidate	Identifies the candidate movement actually evaluated and its scoped release context.
Input and eligibility posture	Current authorization artifacts expose material incompleteness or missing authority honestly; such cases are non-authorizing and fail closed to ESCALATE. Any pre-transition non-formation state is a separate eligibility concept, not a current public decision label.
Decision	The current live public Decision API has one canonical ALLOW, ESCALATE, or BLOCK result; materially incomplete or authority-insufficient inputs do not create a fourth public outcome and must not become autonomous ALLOW.
Evidence and authority	Claims are provenance-bearing and degrade explicitly when unavailable or incomplete.
Release posture	States whether autonomous release is authorized for the scoped movement.
Review posture	When review is required, identifies the unresolved judgment without fabricating reviewer authority or closure.
Consequence evidence	States what is actually known about protected-effect formation and does not infer production prevention from shadow observation.
Integrity and parity	Portable artifacts and human-readable projections reconcile to the same canonical decision truth.
Claim limitation	States the evidence limits of route, enforcement, benchmark, and production claims.

Table 4: Public-safe observable contract for decision authorization. No internal schema, field names, or implementation sequence are implied.

6.1 Non-Fabrication as a First-Class Property

A governance artifact is more trustworthy when it can represent absence honestly. Missing evidence should not become zero evidence. Missing delegated authority should not become authority. A pending reviewer outcome should not become approval. A controlled event should not become a production enforcement claim. An unavailable causal link should not become causal attribution. Stronger claims should appear only when stronger evidence is attached.

7 Current Private Implementation Snapshot: Bounded Evidence Only

The evidence reported here is deliberately narrow. It establishes that a private OntoGuard implementation can generate internally consistent decision-authorization artifacts in a controlled corridor. It does not establish public benchmark superiority, population-level performance, generalization across all domains, legal correctness, production route completeness, or deployment-specific non-bypassability.

7.1 Five-Case Controlled Batch

A controlled batch executed on 5 September 2026 contained five independently governed cases chosen to exercise distinct authorization outcomes. Table 5 reports only the public-safe scenario family and observed decision. It does not disclose prompts, trigger conditions, thresholds, expected-output logic, reason-generation logic, or decision mechanisms.

For this archived controlled engineering snapshot only, the same exact batch package was evaluated by the then-unchanged engineering conformance harness, which reported 2,501 passing assertions, zero warnings, and zero hard failures. The count is a descriptive property of that frozen run and

Controlled scenario family	Observed decision	What this single run demonstrates
Low-risk internal support	ALLOW	The implementation can authorize a scoped movement rather than defaulting to review or refusal.
Financial decision support	ESCALATE	A consequential case can be withheld for additional judgment without being collapsed into BLOCK.
Healthcare decision support	ESCALATE	Review routing can remain distinct from autonomous authorization.
Legal / contract decision support	ESCALATE	Multiple domains can reach the same bounded outcome for different case evidence without implying a fixed domain rule.
Unsafe high-impact instruction	BLOCK	A formed movement can be represented as inadmissible for autonomous release.

Table 5: Observed outcomes in one controlled batch. The rows are evidence of execution, not general policy rules or prevalence estimates.

artifact set. It is not a standing score, cumulative product metric, rating, benchmark sample count, or promise that future harness revisions will report the same number. Its only scientific use here is to document that the configured conformance checks for that snapshot completed without warnings or hard failures.

7.2 Artifact-Conformance Evidence

The controlled run exposed several public-safe conformance properties without requiring disclosure of the private implementation:

Observed property	Public interpretation
All three bounded decision classes occurred	The implementation is not restricted to a single fixed output class in the controlled set.
Case-level buyer/audit decision parity	Independent artifact views preserved the same case decision.
Governed supplied-output binding	The decision remained associated with the supplied output being evaluated.
Fixed portable artifact bundle	The implementation produced a bounded machine- and human-readable governance package rather than an open-ended debug dump.
Schema and integrity checks passed	The exact controlled package satisfied its defined structural and integrity assertions.
No harness warnings or hard failures	The unchanged conformance suite found no configured acceptance failure in that run.

Table 6: Public-safe interpretation of the controlled engineering evidence.

Supported by this controlled snapshot	Not established by this snapshot
All three bounded outcomes can occur in one controlled batch.	Population-level accuracy, sensitivity, specificity, calibration, or benchmark superiority.
Case-level artifact decision parity can be verified.	Legal compliance or professional correctness for arbitrary real-world cases.
The governed supplied output can remain bound to the case decision.	Complete observability of every external agent action or hidden reasoning.
Withheld movements can carry explicit non-release semantics.	Production non-bypassability across an unverified customer topology.
A fixed portable artifact bundle can pass an unchanged conformance harness.	Security or correctness of every deployment, integration, adversarial route, or domain.

Table 7: Bounded interpretation of the September 2026 controlled implementation evidence.

7.3 What the Snapshot Does and Does Not Establish

8 Evaluation Methodology

The public scientific claims are intentionally testable without access to the private implementation. A decision-authorization system should be evaluated against observable outcomes and evidence contracts rather than proprietary method details.

8.1 Research Questions

A rigorous evaluation should ask:

1. Does decision authorization reduce unsafe autonomous release relative to systems that have generation or tool permissioning but no semantic authorization boundary?
2. Does the present system fail closed on materially incomplete or authority-insufficient authorization inputs, withholding autonomous release through ESCALATEwhile preserving honest missing-evidence and missing-authority posture?
3. Does it preserve the identity of the exact candidate movement across decision, handoff, and artifact surfaces?
4. Does it improve the precision with which evidence and authority are tied to the specific release decision?
5. Can it route ambiguous or authority-incomplete cases to review without converting uncertainty into unsupported authorization?
6. Can independent evaluators reproduce consequence/no-bind claims from controlled evidence?
7. Do materially changed governing conditions lead to fresh authorization rather than stale decision reuse?
8. Can longitudinal observations remain descriptive and non-decisioning while still identifying useful governance patterns?
9. Can verified later outcomes generate future-governance recommendations without rewriting historical decisions or autonomously modifying current authority?

8.2 Baselines

Evaluation should compare the framework against multiple baselines:

- a base LLM without retrieval;

- conventional dense RAG;
- graph- or hybrid-retrieval systems;
- a tool-capable agent without a semantic authorization boundary;
- a deterministic permission layer without semantic institutional admissibility; and
- controlled ablations that remove one public responsibility at a time, such as current candidate binding, explicit authority provenance, consequence proof, or review routing.

The relevant alternative is therefore not merely “another RAG system.” Decision authorization addresses a different control boundary.

8.3 Datasets and Scenario Construction

A public benchmark should combine:

- truthfulness and hallucination benchmarks such as TruthfulQA and HaluEval [12, 13];
- tool and agent interaction tasks inspired by AgentBench, ToolEmu, AgentDojo, InjecAgent, and tau-bench [6–10];
- consequential-action simulations with expert-labeled authorization outcomes;
- matched-action pairs in which the technical action stays constant while evidence, authority, release surface, or institutional context changes;
- adversarial identity, stale-context, wrong-subject, and mismatched-authorization tests; and
- post-hoc trace sets that explicitly vary observability so the system can be tested for over-claiming prevention and causality discipline.

8.4 Primary Metrics

Public-safe, implementation-independent metrics include:

- decision-routing accuracy and macro-F1 against expert labels;
- unsafe-release false-negative rate;
- unnecessary-escalation rate;
- unsafe-allow rate on materially incomplete or authority-insufficient authorization cases;
- candidate/decision identity-binding accuracy;
- evidence-to-decision attribution precision and recall;
- authority-status accuracy, including correct recognition of missing authority;
- consequence/no-bind proof correctness against controlled ground truth;
- stale-authorization rejection rate under changed conditions;
- cross-artifact decision parity and integrity failure rate;
- calibration metrics, where confidence is exposed, such as expected calibration error and Brier score;
- reviewer time-to-decision and reviewer agreement; and
- mean and tail latency for the full authorization path.

8.5 Claim-to-Test Matrix

Table 8 maps the paper’s central claims to public tests and observable failure modes.

Claim	Public test	Observable failure
Semantic context matters	Matched-action pairs with expert labels and changed institutional context	Decisions ignore material context or vary without label-relevant change.
Semantic authorization adds value	Compare identical agent/control stack with and without the semantic boundary	No reduction in unsafe release, or gains arise only by blocking or escalating nearly everything.
Authorization is movement-bound	Replay authorization against changed candidate, subject, route, or context	Stale or mismatched authorization remains accepted.
Insufficiency fails closed	Construct materially incomplete and authority-insufficient authorization cases	System emits autonomous ALLOW, hides missing evidence/authority, or fails to withhold release through the current ESCALATE path.
Explanation is non-authoritative	Change presentation/projection inputs while holding canonical case evidence fixed	Current-event decision changes because of explanation or display layer.
No-bind claims are evidence-bound	Compare claimed consequence state with controlled commit/no-commit ground truth	System claims prevention or non-formation without supporting consequence evidence.
Longitudinal analytics are descriptive	Append comparable history while holding current-event canonical evidence fixed	Historical analytics silently rewrite the current-event decision.
Later outcomes are prospective	Attach later reviewer/business outcome after issuance	Historical decision mutates or new current authority appears without fresh authorization.

Table 8: Claim-to-test matrix. Each central scientific claim has an observable failure mode.

8.6 Longitudinal Evaluation

Longitudinal governance requires separate safeguards. Batch membership must not be treated as temporal history. Episodes should be linked only when subject/task lineage is authoritative and comparable. Analyses should report when history is insufficient or non-comparable. Pattern claims should remain observational unless an experimental or quasi-experimental design supports causal interpretation.

A useful longitudinal evaluation should measure whether the system detects repeated governance patterns while preserving current-event decision invariance. It should also test whether historical artifacts remain immutable after later outcomes are attached and whether any future-governance recommendation remains inactive until independently authorized.

8.7 Reproducibility Without Source Disclosure

A strong public evaluation does not require publication of the private algorithm. A reproducible benchmark release can instead include:

- public or safely synthetic scenario inputs and release contexts;
- expert-labeling instructions, adjudication protocol, and inter-rater agreement;
- baseline system configurations and fixed model/version identifiers;
- the observable decision-authorization outputs needed for scoring;
- controlled consequence ground truth for no-bind and release tests;
- evaluation scripts, metrics, preregistered exclusions, and statistical procedures; and
- a stable public version identifier or digest for the evaluated private implementation.

This allows independent verification of claims such as context sensitivity, stale-authorization rejection, identity preservation, decision parity, and unsafe-release reduction without revealing the implementation mechanism.

8.8 Statistical Reporting

Public evaluation should report confidence intervals and sensitivity to scenario composition, model choice, domain, and annotation disagreement. Paired bootstrap, stratified permutation tests, or hierarchical models can be used depending on the benchmark structure. Reported gains should separate accuracy improvements from risk-routing tradeoffs: a system that blocks everything may achieve a low unsafe-release false-negative rate while being operationally unusable.

9 Falsifiability and Claim Boundaries

The framework is scientifically useful only if its claims can fail. The following observations would materially weaken or falsify the central thesis:

- semantic decision authorization does not reduce unsafe release relative to simpler permission or review baselines;
- the system cannot reliably preserve the identity of the movement it claims to govern;
- materially changed context does not produce correspondingly fresh evaluation when expert labels require it;
- authority or evidence claims routinely overstate what is attached;
- explanation or presentation layers can alter the canonical decision;
- stale or materially mismatched authorization remains accepted as current;
- controlled no-bind claims cannot be reproduced from independent consequence evidence;
- longitudinal analytics affect current decisions despite being presented as descriptive; or
- later feedback can alter current or historical authority without explicit prospective authorization.

Governance artifacts can look persuasive even when their underlying authority is weak. A scientifically credible system must therefore make its own insufficiencies and claim boundaries observable.

10 Threats to Validity

10.1 Construct Validity

Institutional admissibility is not a universal natural label. Expert judgments can differ across organizations, release surfaces, jurisdictions, and delegated-authority structures. Public benchmarks should therefore separate broad safety labels from organization-specific authorization labels and report adjudication disagreement rather than treating one annotator as unquestionable ground truth.

10.2 Internal Validity

Scenario authors can accidentally encode obvious lexical cues that make a benchmark easier than the real problem. Matched-action pairs, counterbalanced wording, blinded evaluators, and

ablations are needed to show that a system responds to governing conditions rather than superficial tokens. Private implementation tuning must also be separated from held-out evaluation.

10.3 External Validity

A controlled corridor cannot establish generalization to every enterprise workflow, model, agent framework, or enforcement topology. Public evaluation should therefore vary model families, task domains, consequence classes, and deployment structures before broader claims are made.

10.4 Measurement and Observability

No-bind and longitudinal claims are only as strong as the evidence about actual downstream behavior. Missing trace coverage, ambiguous subject identity, or unavailable outcomes should reduce the strength of the claim rather than be imputed as success.

10.5 Private-Implementation Reproducibility

Because the private mechanism is not disclosed, independent researchers cannot reproduce the internal algorithm from this paper. The appropriate reproducibility target is the observable contract and benchmark behavior. This is a deliberate limitation and should remain explicit until additional implementation disclosure is compatible with the intellectual-property strategy.

11 Security, Ethics, and Institutional Boundaries

Decision authorization should not be presented as a substitute for legal advice, medical judgment, financial supervision, safety engineering, or accountable organizational authority. It is infrastructure for making an AI-mediated movement explicit, evaluating admissibility under attached conditions, routing unresolved judgment, and preserving evidence about decision and consequence.

Several constraints follow:

1. **No false compliance.** Regulatory or policy alignment is not itself a legal conclusion.
2. **No fabricated authority.** Missing approval, reviewer qualification, delegated authority, or policy ownership must remain missing.
3. **No unsupported enforcement claim.** A semantic decision does not prove production non-bypassability unless the deployment’s enforcement topology has been independently demonstrated.
4. **No hidden-reasoning requirement.** Observable agent behavior can be governed without collecting or inferring private chain-of-thought.
5. **No autonomous self-modification.** Descriptive longitudinal signals and feedback recommendations must not silently rewrite current policy, models, authority, or historical decisions.
6. **No causality by correlation.** Behavioral trends are descriptive unless stronger evidence supports a causal statement.
7. **No fabricated economic value.** Commercial value claims require customer-specific baselines and assumptions rather than invented savings.

12 Limitations

This work has important limitations. First, the controlled implementation snapshot contains only five cases and is not a public benchmark. Second, the private implementation prevents reproduction of method internals; this paper supports observable-contract reproducibility instead. Third, institutional admissibility depends on the quality and authority of supplied context. Fourth, external enforcement remains deployment-specific. Fifth, longitudinal analysis requires comparable history and attributable outcomes; sparse or ambiguous history must remain insufficient. Sixth, ontology-mediated systems can inherit ontology gaps or representational bias. Finally, broader comparative evaluation against agent, policy, and assurance baselines remains future work.

13 Future Work

The immediate scientific agenda is evaluation-oriented:

- release an expert-labeled decision-authorization benchmark built from public or safely synthetic consequential-action scenarios;
- preregister matched-action changed-context experiments for the institutional-admissibility hypothesis;
- compare semantic authorization against RAG-only, agent-only, deterministic permission, and review-only baselines;
- study stale-authorization rejection and movement-identity preservation under adversarial mutation;
- evaluate human-review usefulness, reviewer agreement, and evidence-sufficiency judgments;
- develop public conformance tests for semantic-decision handoff without publishing proprietary handoff construction;
- evaluate bounded trace governance under partial observability and adversarially incomplete traces;
- study longitudinal governance as descriptive portfolio intelligence while preserving strict separation from current-event decision authority; and
- study review-gated future-governance feedback under explicit prospective authority with immutable historical records.

14 Conclusion

The transition from generative AI to agentic and tool-mediated systems changes the governance problem. It is no longer enough to ask whether a model can retrieve relevant evidence, generate a plausible answer, or invoke an available tool. Institutions also need to determine whether a particular AI-mediated movement may legitimately become a protected consequence under the conditions that govern it now.

This paper names that problem *decision authorization* and presents a public-safe systems framework for semantic institutional admissibility. The framework separates capability from structural permission, current fail-closed authorization from the separate pre-transition formation research question, the semantic decision responsibility from deployment enforcement responsibility at a conceptual level, and decision from consequence proof. It further places event-level authorization

within a bounded research program for observable behavioral governance and prospectively authorized improvement.

A private OntoGuard implementation provides narrow evidence that the outward contract is executable: a five-case controlled batch produced all three decision classes and passed an unchanged engineering conformance harness with no warnings or hard failures. That result is an implementation milestone, not a claim of benchmark superiority or production non-bypassability.

The durable scientific claim is both narrower and more consequential: **technical capability and structural permission are not equivalent to institutional admissibility**. A trustworthy AI control stack needs a place where that distinction can be evaluated for the exact movement, bound to current evidence and authority, made available for deployment enforcement consumption where applicable, and proven no more strongly than the evidence permits.

A Public Disclosure Boundary

Table 9 summarizes the disclosure policy used for this manuscript. The exclusions are deliberate and should not be interpreted as absent implementation capability.

Disclosed	Intentionally not disclosed
Decision-authorization problem definition	Proprietary decision algorithms, formulas, thresholds, weights, and precedence
Semantic institutional admissibility distinction	Exact formation-eligibility, authority-resolution, and current-validity mechanics
Bounded ALLOW/ESCALATE/BLOCK outward semantics	Internal rule libraries, reason-generation logic, and release mappings
High-level evidence, authority, review, and consequence responsibilities	Private ontology, retrieval, evidence-selection, citation-resolution, and arbitration methods
Conceptual separation of semantic decision responsibility and deployment enforcement responsibility	Exact handoff structure, integrity construction, freshness protocol, partner adapter logic, and any deployment-specific component split
Bounded controlled implementation outcomes	Private prompts, fixtures, internal case identifiers, trigger conditions, thresholds, and expected-output logic
Existence of bounded observable trace and longitudinal governance capabilities	Episode schemas, history design, trajectory methods, semantic metrics, pattern detectors, windowing, and portfolio algorithms
Existence of review-gated future-governance feedback	Candidate-generation, counterfactual evaluation, approval, activation, versioning, and revocation mechanisms
Public evaluation questions, metrics, hypotheses, and falsifiers	Source code, storage architecture, lifecycle ordering, serialization, compaction, sealing, rendering, and finalization logic

Table 9: Conservative public disclosure boundary. The manuscript publishes scientific claims and observable contracts, not implementation recipes.

B Interpretation of the Controlled Evidence

The five-case 5 September 2026 run is preserved in this paper as a *controlled engineering snapshot*: a historical observation of one frozen implementation state, exact artifact set, and then-current conformance harness. It should not be reused as evidence for claims it was not designed to test. The conformance assertion count is not a standing score, cumulative quality metric, benchmark sample count, or prevalence estimate for real enterprise traffic, and this manuscript does not silently update that number when later harnesses add or remove assertions.

A future public benchmark should preregister scenario construction, expert labeling, evaluation metrics, baseline systems, statistical tests, claim thresholds, and held-out conditions. Until such a benchmark exists, the present evidence remains a controlled engineering result.

C Intellectual-Property Notice

Certain implementation aspects of OntoGuard are proprietary and may be subject to pending or future intellectual-property protection. This manuscript intentionally limits disclosure to scientific abstraction, externally observable contract, bounded implementation evidence, and evaluation methodology. No inference should be drawn from the manuscript regarding undisclosed implementation details.

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